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Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2026

Regulation: R20 | University: JNTUK

Strength of Materials (21CV31)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Simple Stresses, Strains & Elastic Constants]

- a) Explain Hooke law and derive the relation between the elastic constants.
- b) A composite bar is subjected to axial load - determine the stress in each section.

UNIT - 2 [Shear Force & Bending Moment Diagrams]

- a) Draw the shear force and bending moment diagrams for the given beam.
- b) Draw the shear force and bending moment diagrams for the given beam.

UNIT - 3 [Bending & Shear Stresses]

- a) Derive the bending equation $M/I = \sigma/y = E/R$.
- b) Explain the distribution of shear stress in a rectangular section.

UNIT - 4 [Torsion of Shafts & Springs]

- a) Derive the torsion equation and the power transmitted by a shaft.
- b) Derive the torsion equation and the power transmitted by a shaft.

UNIT - 5 [Principal Stresses, Mohr Circle & Columns]

- a) Explain Mohr circle for the determination of principal stresses.
- b) State and explain Euler theory of columns.